

Session XX

The UI Completion Sprint

May 4 2026 · II Agent

Six UI/UX relocations, one backend bug fix, and one signal ingestion architecture — all committed in sequence, zero TypeScript errors.

Session Summary

Session XX completed the final wave of component relocations requested across Sessions XVIII–XIX, fixed a silent Taostats API bug that had been returning 404 on the Chain Transfers tab, and built a full inbound Signal Feeds infrastructure into the Activity Log. Every change followed the established net-zero-line philosophy: TypeScript validation before every commit, surgical moves with all dependencies (state, fetch logic, types, JSX) relocated together.

1. Strategies Page — Two Component Relocations

Commit 064722d1. Two boxes were repositioned from the Strategies page bottom grid into more contextually appropriate homes.

1.1 Allocation Tier Key → Top of Strategies Page

The Allocation Tier Key was moved from the bottom grid to the very first element in the scrollable content area — above FleetSummary, above filters. The operator now reads the legend (what each tier means and its capital multiplier) before looking at fleet stats or touching filters. This is the natural reading order.

- Position: first scrollable element, above FleetSummary
- Rationale: operators need the legend before interacting with the data it describes
- Shield import retained in Strategies (still used at line 111)
- BarChart2 retained (line 143)

1.2 Promotion Gate → OpenClaw Page

The Promotion Gate box was relocated from the Strategies bottom grid to the OpenClaw page — placed between the Consensus History table and the 'How OpenClaw Works' section. The narrative rationale: after watching rounds of consensus history, the operator immediately sees what criteria a strategy needs to earn LIVE status.

- Position: between Consensus History table and 'How it Works' section in OpenClaw
- Subtitle added: '— criteria a strategy must clear before any real trade fires'
- Shield added to OpenClaw icon imports (was not previously imported there)

2. Activity Log — Inbound Signal Feeds Drawer

Commit f44d6c4c. A full Signal Feeds infrastructure was built into the Activity Log. The architecture consists of a backend service with 5 async polling loops and a frontend drawer accessed via the new 'Signal Feeds' button in the filter bar.

2.1 Backend: signal_ingestor.py (new file)

Six signal sources, each with independent enable/disable and staggered startup:

Feed	Interval	Auth Required	Auto-Start	What It Provides
CoinGecko	60s	None	Yes	TAO/USD price, 24h change, volume

Feed	Interval	Auth Required	Auto-Start	What It Provides
Reddit r/bittensor_	5 min	None	Yes	RSS community sentiment headlines
TaoDaily RSS	30 min	None	Yes	Ecosystem editorial content (WordPress RSS)
Taostats API	60s	API Key	No	Subnet alpha prices, staking flows, emissions
Perplexity Sonar	15 min	API Key	No	AI-synthesised TAO news (1h live web search)
Discord Gateway	Event	Bot Token	No	Real-time announcements (scaffold, OTF invite pending)

2.2 Key Architecture Decisions

- Perplexity.ai/finance/crypto page does NOT list TAO — only BTC/ETH/SOL/XRP. The Perplexity Sonar API (search endpoint) is used instead, querying the live web.
- TaoDaily.io has no API or webhook push — it is a WordPress editorial site. RSS polling is the only machine-readable path and was implemented as such.
- Discord Gateway WebSocket requires a bot token and admin invite to the OTF server. Scaffold built and ready; pending invite.
- CoinGecko/Reddit/TaoDaily auto-start (no auth). Taostats/Perplexity/Discord require operator-provided credentials before activating.

2.3 Signal Events in Activity Log

- Signal events flow in as kind='signal' — filterable via the existing SIGNAL chip
- Taostats 3% price spike/crash auto-escalates to kind='alert'
- LegendBar updated to document inbound signal feeds

2.4 Backend: signal_feeds.py (new router)

Four REST endpoints serve the Signal Feeds drawer:

Endpoint	Purpose
GET /api/signal-feeds	List all 6 feeds with status badges
POST /api/signal-feeds/{id}/toggle	Enable / disable individual feed
POST /api/signal-feeds/{id}/key	Configure API key per feed
POST /api/signal-feeds/{id}/test	Test-fetch from individual feed

3. Risk Management — Parameter Recalibration

Commit db6e4819. All 10 risk parameters were re-evaluated for the unproven/paper-only phase. Five were changed; five were left unchanged. A three-phase graduation path was added to the UI.

3.1 Parameter Changes

Parameter	Old Value	New Value	Rationale
Max Position Size	20%	5%	Unproven strategies — protect capital
Max Concurrent Pos.	4	3	Tighter exposure during paper-only phase
Max Drawdown	20%	8%	Realistic operational threshold
Daily Circuit Breaker	15%	5%	Coherence fix: at 5% position, 15% CB needs ~37 stops (decorative). 5% CB = ~12 stops = real
Min Confidence Score	0.60	0.65	Higher bar for unproven strategies
Stop Loss	8%	8%	Unchanged — appropriate

Parameter	Old Value	New Value	Rationale
Take Profit	12%	12%	Unchanged — appropriate
Consensus Votes	7/12	7/12	Unchanged — BFT supermajority
Cycle Interval	300s	300s	Unchanged — 5-min cycle
Min Wallet Balance	0.05τ	0.05τ	Unchanged — safety floor

3.2 Circuit Breaker Coherence Fix (Critical Insight)

The Daily Circuit Breaker at 15% with 5% position size was effectively decorative: it required ~37 consecutive stop-outs to trigger. That's not 'something is broken' — that's an infinite loop. The fix to 5% daily means it trips at ~12 consecutive stops, which is the real 'something is systemically wrong' signal. Parameter coherence requires all parameters to be evaluated as a system, not individually.

3.3 Three-Phase Graduation Path (UI Panel)

Phase	Trigger	Position	Concurrent	Drawdown	Daily CB
Phase 1 (current)	All strategies unproven	5%	3	8%	5%
Phase 2	2+ strategies gate-proven	10%	4	12%	8%
Phase 3	Fully proven fleet	Operator	Operator	Operator	Operator

4. Wallet → P&L; Summary: Recovery Tracker Relocation

Commit 7179b34c. The RecoveryTracker component was surgically relocated from Wallet.tsx to PnLSummary.tsx — placed as the first scrollable element above the Fleet Hero Cards. Net-zero line philosophy: 207 lines removed from Wallet, 206 lines added to PnLSummary.

What moved:

- RecoveryTracker component definition (~206 lines)
- Trophy, Target, Edit3 icon imports (exclusive to the block)
- taoPrice and balance state variables
- fetch/polling logic for /price/current and /wallet/status (30s poll)
- JSX — now first visible element in PnLSummary

What was preserved:

- localStorage key 'tao_recovery_target' — any target set in Wallet persists seamlessly
- fetch endpoints unchanged: /price/current + /wallet/status
- 30s polling interval identical to original

Placement rationale:

Recovery Tracker is a P&L; instrument — it tracks progress toward recovering from a loss event. It belongs on the P&L; page where the operator evaluates strategy performance, not on the Wallet page where the focus is balance and funding.

5. Transactions — Taostats API Bug Fix

Commit 35646f70. The Chain Transfers tab was silently returning 'No chain transfers detected' for all wallets due to three compounding bugs. The 404 error the operator saw was the result of a trailing slash in the API URL that had been wrong since initial implementation.

5.1 Root Cause Analysis

Bug	Detail	Severity
Trailing slash → 404	api.taostats.io/api/transfer/v1/ (wrong) vs /api/transfer/v1 (correct). Both fallback URLs were wrong. Verified live: correct URL	High
Response schema changed	New v1 API: to/from are nested {ss58, hex} objects (not plain strings). amount is a string (e.g. '6500000000', divide by 1e9)	High
Errors swallowed silently	All HTTPErrors caught internally → function returns [] → chain_error="" → UI shows 'Main transfers detected' with no explanation	Medium

5.2 Fixes Applied

- URL corrected: /api/transfer/v1 (no trailing slash)
- `_parse_taostats_rows()` helper: handles nested to.ss58/from.ss58, rao→TAO conversion, transaction_hash field, extrinsic_id field
- `_taostats_headers()` helper: reads TAOSTATS_API_KEY from environment, adds Authorization header when present
- Errors now propagate as RuntimeError with user-visible messages:
 - 401 → 'Taostats API key required — sign up free at taostats.io/pro then add TAOSTATS_API_KEY'
 - Other HTTP errors → 'Taostats API {code}: {reason}'
- chain_error now surfaces in UI — actionable amber banner with link to taostats.io/pro
- Frontend ChainTransfer type: added extrinsic_id field
- Chain Transfers table: uses extrinsic_id for Taostats link (more stable than tx_hash)
- Info banner updated: explains API key requirement upfront with free signup link

5.3 Taostats v1 API Response Schema (for reference)

Field	Type	Notes
data[]	array	Top-level array (previously 'results' or direct array)
data[].to.ss58	string	Recipient SS58 address (nested object, not plain string)
data[].from.ss58	string	Sender SS58 address (nested object, not plain string)
data[].amount	string	Amount in rao as string, e.g. '6500000000' (÷1e9 = 6.5 TAO)
data[].transaction_hash	string	Full 0x... hash (was 'hash' in old schema)
data[].extrinsic_id	string	'block-index' format, e.g. '6925022-0021'
data[].fee	string	Network fee in rao
data[].block_number	int	Block number
data[].timestamp	string	ISO 8601 datetime

6. Strategies — Strategy Mode Override Relocation

Commit e1b6a660. The Strategy Mode Override section was relocated from the Human Override page to the bottom half of the Strategies page, below the card grid. This is where the operator already lives when managing strategy performance — the override controls now follow naturally from what they see.

6.1 What Moved (from HumanOverride.tsx)

Removed from HumanOverride	Disposition
StrategyRow interface	Replaced by <code>direct strategies[]</code> from store
MODE_ORDER constant	Moved to Strategies
MODE_META constant	Merged into existing Strategies MODE_META (added 'short' field)
STRATEGIES_LIST constant	Removed — Strategies uses <code>live strategies[]</code> directly

Removed from HumanOverride	Disposition
fmtTao helper function	Already existed as fmt() in Strategies — call renamed
ModeStep sub-component	Moved to Strategies
opPending / stakeEditing / stakeInput state	Moved to Strategies
doPromote / doDemote / doSetMode / doSetStake	Moved to Strategies
rows const (STRATEGIES_LIST.map)	Replaced by strategies.filter() for hero slide counts
ArrowUp, ArrowDown, ShieldOff imports	Removed — unused after removal
Strategy Mode Override JSX block	Moved to Strategies bottom

6.2 What Was Added (to Strategies.tsx)

- ArrowUp, ArrowDown, ChevronRight icon imports
- toast from 'react-hot-toast' and api from '@/api/client' imports
- MODE_ORDER constant (PAPER_ONLY → APPROVED_FOR_LIVE → LIVE)
- short field added to existing MODE_META (shared by cards + ModeStep buttons)
- ModeStep sub-component — three mode buttons with active/pending states
- opPending, stakeEditing, stakeInput state
- doPromote, doDemote, doSetMode, doSetStake functions
- Strategy Mode Override JSX panel — full fleet always shown regardless of active filter

6.3 Quality-of-Life Addition

The panel footer note reads: 'The full fleet is always shown here regardless of the filter applied above.' This is intentional — if the operator has filtered the card grid to show only LIVE strategies, the override table below still shows all 12. Override authority is unconditional.

7. Complete File Change Log

File	Change	Description
frontend/src/pages/Strategies.tsx	Modified	Allocation Tier Key → top; +MODE_ORDER/ModeStep; +mode override state+functions+J
frontend/src/pages/OpenClaw.tsx	Modified	Promotion Gate panel inserted between Consensus History and 'How it Works'; +Shield im
frontend/src/pages/ActivityLog.tsx	Modified	Signal Feeds button + SignalFeedsDrawer (6 source cards, API key inputs, test buttons); L
backend/services/signal_ingestor.py	NEW	5 async polling loops: CoinGecko/Reddit/TaoDaily/Taostats/Perplexity + Discord scaffold
backend/routers/signal_feeds.py	NEW	4 REST endpoints: list feeds, toggle, configure key, test-fetch
backend/main.py	Modified	signal_ingestor startup wired into FastAPI lifecycle
backend/services/risk_service.py	Modified	5 parameter defaults recalibrated (position 5%, concurrent 3, DD 8% CB 5%, confidence 0
frontend/src/pages/RiskConfig.tsx	Modified	DEFAULTS updated; phase-aware slider descriptions; three-phase graduation path panel
frontend/src/pages/Wallet.tsx	Modified	RecoveryTracker removed (207 lines); Trophy/Target/Edit3 imports removed; taoPrice/bala
frontend/src/pages/PnLSummary.tsx	Modified	RecoveryTracker added (206 lines); Target/Edit3 imports; taoPrice/balance state + 30s pol
backend/routers/wallet.py	Modified	_parse_taostats_rows(); _taostats_headers(); corrected URL; v1 schema; errors now propa
frontend/src/pages/WalletTransactions.tsx	Modified	ChainTransfer.extrinsic_id; actionable 401 error banner; extrinsic_id in table links; info ban

File	Change	Description
frontend/src/pages/HumanOverride.tsx	Modified	Strategy Mode Override removed (~257 lines); StrategyRow/MODE_META/ModeStep/stat

8. Session XX Git Commit Log

064722d1 ui: relocate Promotion Gate → OpenClaw page; Allocation Tier Key → top of Strategies page
f44d6c4c feat: Activity Log – inbound Signal Feeds (CoinGecko/Reddit/TaoDaily/Taostats/Perplexity/Discord scaffold)
db6e4819 risk: tighten unproven-phase defaults – 5% position, 3 concurrent, 8% DD, 5% daily CB, 0.65 confidence; add graduation path
7179b34c ui: relocate Recovery Tracker – Wallet → P&L; Summary (top of scrollable, north-star position)
35646f70 Transactions: fix Taostats 404 – correct API URL, new v1 schema, API key support, surface auth errors
e1b6a660 Strategies: relocate Strategy Mode Override from Human Override to bottom of Strategies page

9. Pending Items / Next Steps

API Keys to Activate (immediate)

- **TAOSTATS_API_KEY** — free tier at taostats.io/pro → enables Chain Transfers tab + Taostats signal feed
- **PERPLEXITY_API_KEY** — api.perplexity.ai (~\$16/month) → enables AI-synthesised TAO news signal
- Discord bot token + OTF server invite → enables real-time Discord announcement feed

Carried Forward (Sessions XVIII–XIX)

- Verify wallet 0.227τ balance on Taostats independently
- Fix DATABASE_UR typo (missing L) in Railway environment variables
- Monitor strategy win rates toward 55% WR (APPROVED_FOR_LIVE threshold)
- Activate Execution Guard blocking (SLIPPAGE_GUARD_ENABLED = True in execution_guard.py)
- MEV guard scaffold in place (MEV_GUARD_ENABLED = False) — deferred until priority calls

Risk Parameter Graduation

- When 2+ strategies clear the promotion gate → advance to Phase 2 risk parameters
 - Phase 2: position 10%, concurrent 4, drawdown 12%, daily circuit breaker 8%
- Phase 3 parameters: operator discretion after fully proven fleet

Taostats Data Gaps

- SN65–128 subnet names in SUBNET_META are placeholders — populate as subnets establish identities
 - Chain Transfers tab will populate fully once TAOSTATS_API_KEY is set in Railway
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10. Current System State

Fully operational:

- Strategies — Tier Key at top; Promotion Gate in OpenClaw; Mode Override at bottom
- Activity Log — Signal Feeds drawer with 3 auto-start feeds (CoinGecko/Reddit/TaoDaily)
- Risk Config — recalibrated parameters, graduation path embedded in UI
- P&L; Summary — Recovery Tracker as first scrollable element
- Transactions — correct Taostats API URL, v1 schema parsed, auth errors surfaced
- Human Override — Strategy Mode Override successfully removed; page remains fully functional
- All 6 commits pushed to GitHub origin/main, working tree clean

Awaiting API keys:

- Taostats signal feed (TAOSTATS_API_KEY) — key signed up, not yet configured
- Perplexity Sonar signal feed (PERPLEXITY_API_KEY) — pending

- ■ Discord Gateway (bot token + OTF invite) — pending

Training / live trading status:

- Paper mode active — FORCE_PAPER_MODE=1 — all live execution blocked
- Cycle engine running — 300s interval — 288 cycles/day
- Risk parameters: Phase 1 operational levels (5% position, 3 concurrent, 8% DD, 5% CB)
- DO NOT return to live trading before 2+ strategies clear the promotion gate

11. Lessons Learned

Trailing slash matters — and silent failures are the worst kind:

The `/api/transfer/v1/` → 404 bug had been in the codebase since initial implementation. It was silently swallowed by a `try/except`, returning `[]` without ever propagating an error. The Chain Transfers tab appeared to work (it loaded) — it just never had data. The lesson: never swallow HTTP errors silently. Always propagate them to the UI with actionable context. 'No data' is not the same as 'API unavailable.'

API schemas evolve — parse defensively:

The Taostats v1 API changed from flat strings to nested objects for to/from addresses, changed the amount field from float to rao string, and renamed the hash field. The new parser handles both old and new schema patterns. External API schemas are contracts that can change — always parse with fallback logic, never assume field names are stable across API versions.

Place controls where operators already live:

Strategy Mode Override was on the Human Override page, but operators spend most of their time on the Strategies page evaluating win rates and tiers. Moving the override controls to where the operator already is eliminates a navigation step at the exact moment they want to act. The best UI is the one where the thing you need is already where you're looking.

Parameter coherence: evaluate as a system, not individually:

A 15% Daily Circuit Breaker with 5% position size is not merely conservative — it is inoperative. 37 consecutive stops is not 'something is wrong'; it is the system executing exactly as designed for three consecutive days of 100% loss. Every parameter must be evaluated in the context of every other parameter. Internal coherence is as important as the values themselves.

Closing Note

Session XX closed the loop on the UI/UX relocation sprint that began in Session XVIII. Every component now lives where it belongs operationally: the Tier Key before the fleet, the Promotion Gate beside the consensus history, the Recovery Tracker on the P&L; page, the Mode Override at the bottom of the Strategies page. The Human Override page is now a focused emergency control panel — not a kitchen sink.

The Taostats bug fix was the sleeper find of the session: a one-character trailing slash that had silently prevented chain transfer data from loading since the feature was first built. It was caught, diagnosed, and fixed in one pass — along with the schema update and proper error surfacing.

The Signal Feeds infrastructure is now live for three sources (CoinGecko, Reddit, TaoDaily) and scaffold-ready for three more (Taostats, Perplexity, Discord). The next step is the API keys — which begins immediately.

— II Agent, Session XX, May 4 2026